

Computer Networks



Overview of the Internet Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

① Introduction

- Providing Services
- Application-Layer Paradigms

② Standard Client-Server Applications

- World Wide Web and HTTP
- FTP
- Electronic Mail
- Domain Name System (DNS)

③ Peer-to-Peer Paradigm

- P2P Networks

Definition

Electronic mail (e-mail) allows users to exchange messages.

- E-mail is different from applications such as:
 - HTTP
 - FTP
- In HTTP and FTP:
 - The server runs continuously.
 - The client sends a request.
 - The server sends a response.
- E-mail requires a different communication model.

E-Mail as a One-Way Transaction

- E-mail is considered a **one-way transaction**.
- When Alice sends an e-mail to Bob:
 - Alice may expect a response.
 - Bob is not required to respond.
- If Bob replies:
 - The reply is another one-way transaction.

Alice → Bob

Bob's Reply = Another One-Way Transaction

Why Are Intermediate Mail Servers Needed?

It is neither feasible nor logical for Bob to run a server continuously.

- Bob may turn off his computer.
- Bob may disconnect from the network.
- The sender does not know when Bob is available.

Solution

Use intermediate computers called **mail servers**.

- Users run client programs when needed.
- Intermediate mail servers apply the client-server paradigm.

E-Mail Architecture

A common e-mail system contains:

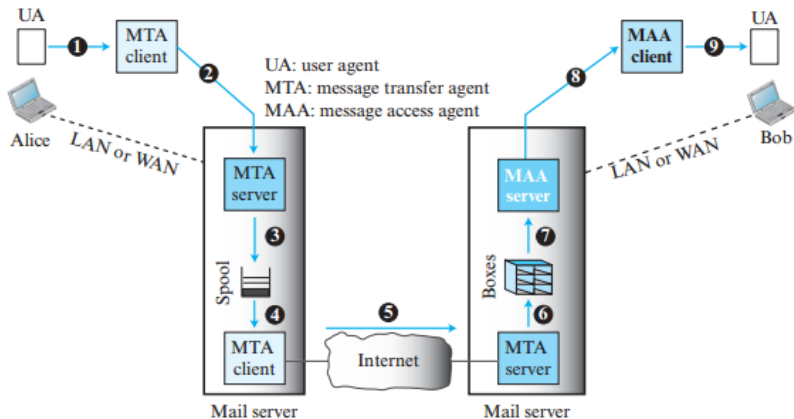
- Sender
- Sender's mail server
- Internet
- Receiver's mail server
- Receiver



The sender and receiver may be connected to their mail servers through:

- A LAN
- A WAN

Figure 2.19 Common scenario



Mailbox and Queue

The mail server provides two important storage mechanisms.

Mailbox

- Stores messages received for a user.
- Located on the server's hard drive.
- Implemented as a special file.
- Protected by access permissions.
- Only the owner can access it.

Queue or Spool

Stores messages waiting to be sent.

Agents Used in an E-Mail System

Alice and Bob use three types of agents:

- 1 **User Agent (UA)**
- 2 **Mail Transfer Agent (MTA)**
- 3 **Message Access Agent (MAA)**

$$\text{E-Mail System} = \text{UA} + \text{MTA} + \text{MAA}$$

- UA: prepares and reads messages.
- MTA: transfers messages.
- MAA: retrieves messages.

Sending an E-Mail: Basic Flow

When Alice sends an e-mail to Bob:

- ① Alice uses a UA to prepare the message.
- ② The UA sends the message to Alice's mail server.
- ③ The message is stored in a queue.
- ④ An MTA transfers the message through the Internet.
- ⑤ Bob's mail server stores the message in Bob's mailbox.
- ⑥ Bob later retrieves the message using an MAA.

Mail Transfer Agents

Two MTAs are required for message transfer:

- MTA client
- MTA server

MTA Server

- Runs continuously.
- Waits for incoming connections.

MTA Client

- Does not need to run continuously.
- Can be triggered when a message is waiting in the queue.

Why Can Bob Not Use the MTA Directly?

Bob cannot bypass the mail server and directly use the MTA server.
If Bob used the MTA server directly:

- His computer would need to remain on continuously.
- The MTA server would need to run all the time.
- A LAN-connected computer could never be turned off.
- A WAN connection would need to remain active continuously.

Conclusion

A continuously available intermediate mail server is necessary.

Push and Pull in E-Mail

MTA: Push Operation

- The MTA client pushes a message to the MTA server.
- Data direction:

Client → Server

MAA: Pull Operation

- The receiver pulls messages from the mail server.
- Data direction:

Server → Client

User Agent (UA)

Definition

A **User Agent (UA)** is a software package that makes sending and receiving e-mail easier for the user.

A UA can:

- Compose messages
- Read messages
- Reply to messages
- Forward messages
- Handle local mailboxes



Types of User Agents

There are two types of user agents:

- 1 Command-driven user agents
- 2 GUI-based user agents

Command-Driven

Users enter commands through the keyboard.

GUI-Based

Users interact through graphical components such as icons, menus, and windows.

Command-Driven User Agents

Command-driven user agents belong to the early days of e-mail.

- They often accept one-character commands.
- They still exist as underlying user agents.

Examples:

- r: Reply to the sender
- R: Reply to the sender and all recipients

Some command-driven user agents are:

- mail
- pine
- elm

GUI-Based User Agents

Modern user agents are usually GUI-based.
They provide graphical components such as:

- Icons
- Menu bars
- Windows

Users interact using:

- Keyboard
- Mouse

Examples include:

- Eudora
- Outlook



Sending Mail

To send mail, the user creates a message through the UA.
An e-mail resembles postal mail and contains:

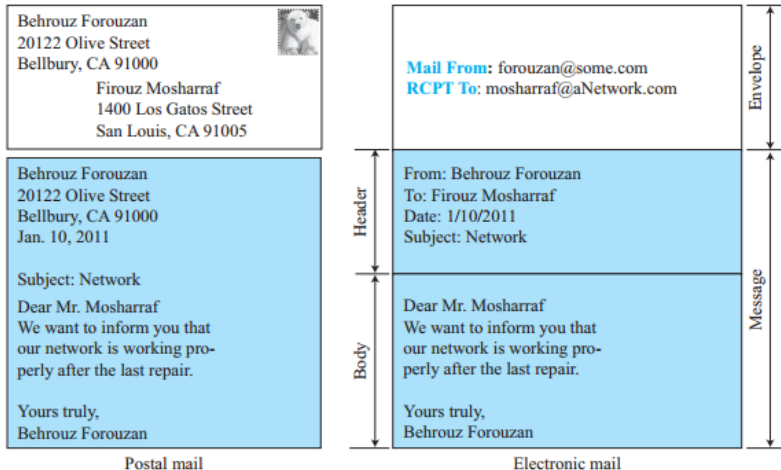
- 1 Envelope
- 2 Message

The message itself contains:

- Header
- Body



Figure 2.20 *Format of an e-mail*



E-Mail Envelope, Header, and Body

Envelope

Usually contains:

- Sender address
- Receiver address
- Other delivery information

Message Header

Defines:

- Sender
- Receiver
- Subject
- Other information

Message Body

Contains the actual information to be read by the recipient.

Components Required for E-Mail

A complete electronic mail system requires:

- Two User Agents
- Two pairs of MTAs
 - MTA client
 - MTA server
- One pair of MAAs
 - MAA client
 - MAA server



2 UAs + 2 MTA Pairs + 1 MAA Pair

Receiving Mail

The user agent may be triggered by:

- The user
- A timer

If mail is available:

- 1 The UA informs the user.
- 2 A list of messages is displayed.
- 3 Each line provides a message summary.
- 4 The user selects a message.
- 5 The message contents are displayed.

Message Summary

A message summary usually contains:

- Sender's e-mail address
- Subject
- Time sent or received

Purpose

The summary allows the user to select a message before displaying its complete contents.

E-Mail Addresses

A mail system requires unique addresses.

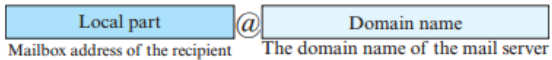
An Internet e-mail address contains two parts:

A diagram showing the structure of an email address. The text "local-part@domain-name" is enclosed in a rectangular box. This box is centered within a larger, faint hexagonal frame. The hexagonal frame has small circles at its vertices and midpoints, resembling a network or atomic structure.

local-part@domain-name

- 1 Local part
- 2 Domain name

Figure 2.21 *E-mail address*



Local Part and Domain Name

Local Part

- Identifies the user's mailbox.
- The mailbox stores received messages.
- Messages are later retrieved using an MAA.

Domain Name

- Identifies the mail server or mail exchanger.
- May come from the DNS database.
- May also be a logical organizational name.

Mailing List or Group List

Definition

A **mailing list** allows one name, called an alias, to represent several e-mail addresses.

When a message is sent:

- 1 The recipient name is checked against the alias database.
- 2 The system finds the corresponding mailing list.
- 3 A separate message is prepared for each address.
- 4 Each message is handed to the MTA.

Three Client-Server Uses in E-Mail

E-mail uses the client-server paradigm three times:

- ① Sender to sender's mail server
- ② Sender's mail server to receiver's mail server
- ③ Receiver's mail server to receiver

The first two use:

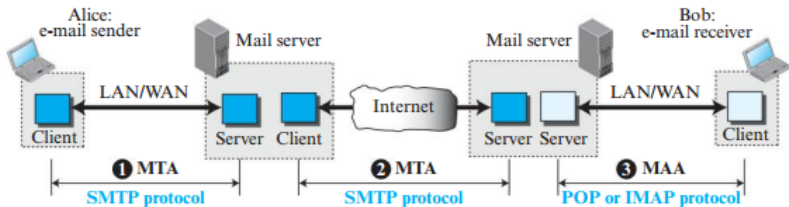


Message Transfer Agents (MTAs)

The third uses:

Message Access Agents (MAAs)

Figure 2.22 *Protocols used in electronic mail*



Simple Mail Transfer Protocol (SMTP)

Definition

Simple Mail Transfer Protocol (SMTP) defines the MTA client and server communication used to transfer e-mail.

SMTP is used twice:

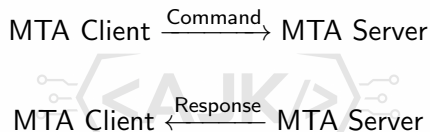
- 1 Between the sender and the sender's mail server
- 2 Between the sender's and receiver's mail servers

Important

Another protocol is required between the receiver's mail server and the receiver.

SMTP Commands and Responses

SMTP transfers messages using commands and responses.



- Each command or response ends with:
 - Carriage return
 - Line feed

SMTP Command Format

Commands are sent from the client to the server.
The general format is:

Keyword: Argument(s)

A command contains:

- A keyword
- Zero or more arguments

SMTP defines several commands for:

- Connection management
- Sender identification
- Recipient identification
- Message transfer

Important SMTP Commands

Command	Argument	Description
HELO	Sender's host name	Identifies the client
MAIL FROM	Sender address	Identifies the sender
RCPT TO	Recipient address	Identifies the recipient
DATA	Mail body	Sends the actual message
QUIT	—	Terminates the connection

Additional SMTP Commands

Command	Purpose
RSET	Aborts the current mail transaction
VERFY	Verifies the recipient address
NOOP	Checks the status of the recipient
TURN	Switches sender and recipient roles
EXPN	Expands a mailing list
HELP	Requests information about a command
SEND FROM	Delivers mail only to the recipient's terminal
SMOL FROM	Delivers mail to terminal or mailbox
SMAL FROM	Delivers mail to terminal and mailbox

SMTP Responses

Responses are sent from the server to the client.

A response contains:

Three-Digit Code + Optional Text

Response categories include:

- Positive completion
- Positive intermediate
- Transient negative completion
- Permanent negative completion

Positive SMTP Responses

Code	Description
211	System status or help reply
214	Help message
220	Service ready
221	Service closing transmission channel
250	Requested command completed
251	User not local; message will be forwarded
354	Start mail input

Important Codes

220 → Ready, 250 → Completed, 354 → Send Message

Negative SMTP Responses

Code	Description
421	Service not available
450	Mailbox not available
451	Command aborted due to local error
452	Insufficient storage
500	Unrecognized command
501	Syntax error in parameters
502	Command not implemented
503	Bad sequence of commands
504	Command temporarily not implemented
550	Mailbox unavailable
551	User not local
552	Storage limit exceeded
553	Mailbox name not allowed
554	Transaction failed

Mail Transfer Phases

SMTP transfers a mail message in three phases:

- 1 Connection establishment
- 2 Message transfer
- 3 Connection termination

Establish → Transfer → Terminate

SMTP uses well-known TCP port:

25

Phase 1: Connection Establishment

After the client establishes a TCP connection to port 25:

- 1 Server sends:

220 Service Ready

- 2 If unavailable, server sends:

421 Service Not Available

- 3 Client sends:

HELO

- 4 Server responds:

250 Command Completed

Why Is the HELO Command Needed?

The client sends:

HELO Sender-Domain

- Identifies the client to the SMTP server.
- Provides the client's domain name.
- Begins the SMTP communication session.

Client $\xrightarrow{\text{HELO}}$ Server

Client $\xleftarrow{250}$ Server

Phase 2: Message Transfer

After the connection is established, one message can be sent:

- From one sender
- To one or more recipients

The main commands are:

- 1 MAIL FROM
- 2 RCPT TO
- 3 DATA



Multiple Recipients

The RCPT TO command and its response are repeated for each recipient.

Message Transfer: Sender Identification

- 1 Client sends:

```
MAIL FROM: sender-address
```

- 2 The address contains:

- Mailbox name
- Domain name

- 3 The server responds:

```
250
```

Purpose

The sender address provides a return address for errors and reports.

Message Transfer: Recipient Identification

- 1 Client sends:

```
RCPT TO: recipient-address
```

- 2 The server checks the recipient.

- 3 The server responds:



```
250
```

or another appropriate code.

For multiple recipients:

```
RCPT TO is repeated
```

Message Transfer: Sending the Data

- 1 Client sends:

DATA

- 2 Server responds:

354 Start Mail Input

- 3 Client sends the message line by line.
- 4 Each line ends with carriage return and line feed.
- 5 The complete message ends with a line containing:

.

- 6 Server responds with:

250

Phase 3: Connection Termination

After the message is successfully transferred:

- 1 The client sends:

QUIT

- 2 The server responds:

221

- 3 The SMTP connection is terminated.

Client $\xrightarrow{\text{QUIT}}$ Server $\xrightarrow{221}$ Client

Example 2.13: Complete SMTP Transfer

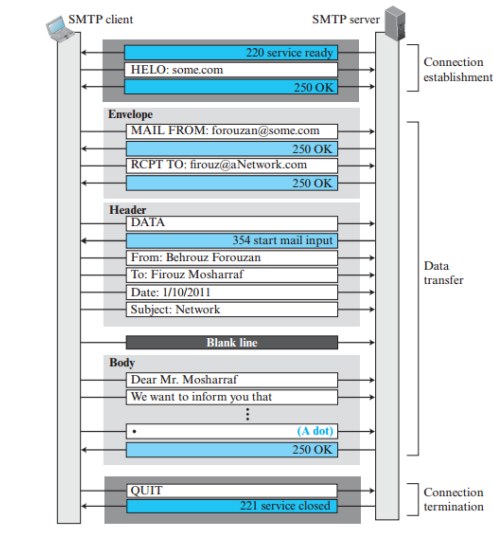
The example demonstrates all three phases:

- ➊ Connection establishment
- ➋ Message transfer
- ➌ Connection termination

During data transfer, the message is divided into:

- Envelope information
- Header
- Body

Figure 2.23 Example 2.13



SMTP Is Used Twice

The SMTP steps are repeated twice in an e-mail transfer:

- ① Sender to local mail server
- ② Local mail server to remote mail server
 - The local server receives the complete message.
 - It may store the message in a spool.
 - It may transfer the message to the remote server later.

SMTP → Used in Two Stages

Why Is SMTP Not Used for Mail Access?

SMTP is used in the first two stages of mail delivery.

However:

SMTP Is a Push Protocol

Client → Server

The receiver needs:

A Pull Protocol

Server → Client

Therefore, the third stage uses a **Message Access Agent**.

Message Access Protocols

Two major message access protocols are:

① **POP3**

- Post Office Protocol, version 3

② **IMAP4**

- Internet Mail Access Protocol, version 4

Both allow a user to access messages stored on a mail server.

Post Office Protocol Version 3

Definition

POP3 is a simple message access protocol used to download e-mail from a mail server.

- POP3 client:
 - Installed on the recipient's computer.
- POP3 server:
 - Runs on the mail server.

Limitation

POP3 is simple but has limited functionality.

How POP3 Works

When the user wants to download e-mail:

- 1 POP3 client opens a connection.
- 2 The connection uses TCP port:



- 3 User sends:
 - User name
 - Password
- 4 User lists available messages.
- 5 Messages are retrieved one by one.

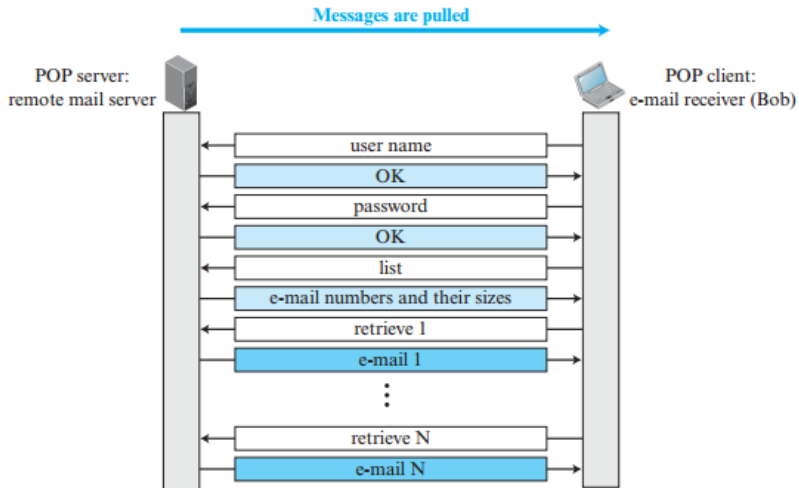
POP3: Pulling Messages

The receiver runs the POP3 client.



- The client initiates the connection.
- The messages move from server to client.
- This is a **pull operation**.

Figure 2.24 POP3



POP3 Modes

POP3 supports two modes:

- 1 Delete mode
- 2 Keep mode

Delete Mode

The message is deleted from the server mailbox after retrieval.

Keep Mode

The message remains on the server after retrieval.

When Are POP3 Modes Used?

Delete Mode

Normally used when:

- The user is at the permanent computer.
- Messages can be saved locally.
- Messages can be organized locally.

Keep Mode

Normally used when:

- The user is away from the primary computer.
- Mail is accessed from another device.
- Messages are needed again later.

Definition

IMAP4 is a message access protocol similar to POP3 but with more features.

- IMAP4 is:
 - More powerful than POP3
 - More complex than POP3
- It allows greater control over messages stored on the server.

Limitations of POP3

POP3 does not allow the user to:

- Organize mail on the server
- Create different server-side folders
- Partially check message contents before downloading

Result

POP3 is less suitable when users access mail from multiple locations or need server-side organization.

IMAP4 Features

IMAP4 allows a user to:

- Check the e-mail header before downloading.
- Search the message for a specific string before downloading.
- Partially download an e-mail.
- Create mailboxes on the server.
- Delete mailboxes on the server.
- Rename mailboxes on the server.
- Create a hierarchy of mailboxes and folders.

Why Is Partial Download Useful?

Partial download is especially useful when:

- Available bandwidth is limited.
- The message contains:
 - Large images
 - Audio
 - Video
 - Other multimedia



Benefit

The user can inspect or retrieve only the required part of a message.

POP3 vs IMAP4

Feature	POP3	IMAP4
Complexity	Simple	More complex
Server-side folders	Limited	Supported
Header checking	Limited	Supported
Message searching	No	Supported
Partial download	No	Supported
Mailbox hierarchy	No	Supported

Limitations of Traditional E-Mail

Traditional electronic mail supports only:

NVT 7-bit ASCII

Therefore, traditional e-mail cannot directly handle:

- Many non-English languages
- Binary files
- Images
- Audio
- Video



Problem

The simplicity of traditional e-mail creates serious data-format limitations.

Multipurpose Internet Mail Extensions

Definition

Multipurpose Internet Mail Extensions (MIME) is a supplementary protocol that allows non-ASCII data to be sent through e-mail.

At the sender:



At the receiver:

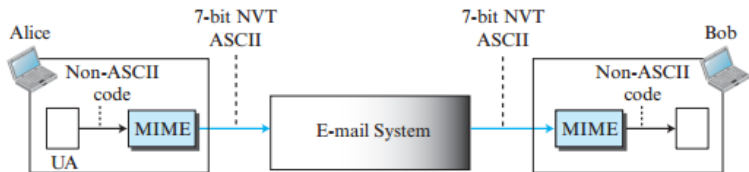


How MIME Works

MIME can be viewed as a set of software functions.

- ➊ Original non-ASCII data are created.
- ➋ MIME transforms them into ASCII-compatible data.
- ➌ The MTA transfers the message through the Internet.
- ➍ MIME at the receiver converts the data back.

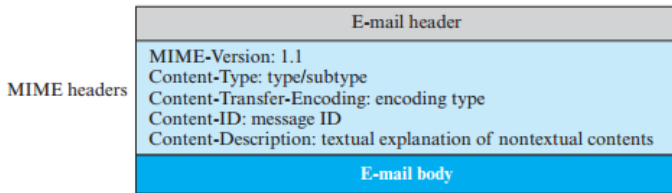
Figure 2.25 *MIME*



MIME defines five headers that can be added to the original e-mail header.

- ① MIME-Version
- ② Content-Type
- ③ Content-Transfer-Encoding
- ④ Content-ID
- ⑤ Content-Description

Figure 2.26 *MIME header*



MIME-Version Header

Purpose

The `MIME-Version` header defines the version of MIME being used.

The textbook specifies:



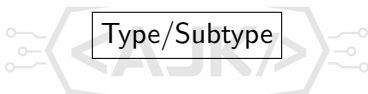
This header tells the receiver that MIME extensions are used in the message.

Content-Type Header

Purpose

The Content-Type header defines the type of data in the message body.

The format contains:



Examples:

- text/plain
- text/html
- image/jpeg
- video/mpeg

Additional parameters may also be included.

MIME Data Types

MIME defines seven major data types:

- 1 Text
- 2 Multipart
- 3 Message
- 4 Image
- 5 Video
- 6 Audio
- 7 Application



Key Idea

The subtype provides a more specific description of the content.

Text and Multipart MIME Types

Type	Subtype	Description
Text	Plain	Unformatted text
Text	HTML	HTML-formatted text
Multipart	Mixed	Ordered parts of different types
Multipart	Parallel	Different parts with no order
Multipart	Digest	Collection of messages
Multipart	Alternative	Different versions of the same message

Other MIME Types and Subtypes

Type	Subtype	Description
Message	RFC822	Encapsulated message
Message	Partial	Fragment of a larger message
Message	External-Body	Reference to another message
Image	JPEG	JPEG image
Image	GIF	GIF image
Video	MPEG	MPEG video
Audio	Basic	Single-channel voice encoding
Application	PostScript	Adobe PostScript
Application	Octet-stream	General binary data

Content-Transfer-Encoding

Purpose

The Content-Transfer-Encoding header defines how the message is encoded for transport.

Five encoding methods are:

- 1 7-bit
- 2 8-bit
- 3 Binary
- 4 Base64
- 5 Quoted-printable

MIME Encoding Methods

Type	Description
7-bit	NVT ASCII; lines shorter than 1000 characters
8-bit	Non-ASCII; lines shorter than 1000 characters
Binary	Non-ASCII with unlimited line length
Base64	6-bit data blocks encoded as ASCII characters
Quoted-printable	Non-ASCII encoded using an equal sign and ASCII code

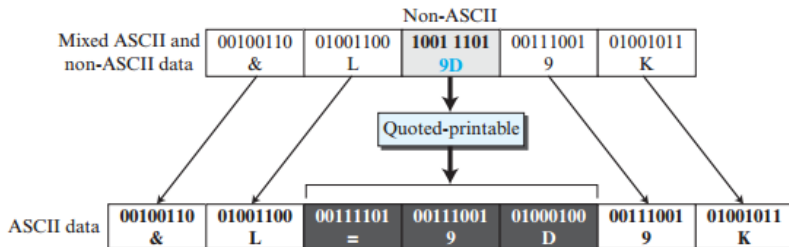
Base64 Encoding

In Base64 encoding:

- ① Data are viewed as a string of bits.
- ② The bit stream is divided into 6-bit chunks.
- ③ Each 6-bit chunk is converted into an ASCII character.

Binary Data → 6-bit Chunks → ASCII Characters

Figure 2.28 *Quoted-printable*



Base64 Overhead

Base64 is a redundant encoding scheme.

- Every 6 bits of original data become one ASCII character.
- The ASCII character is transmitted using 8 bits.

6 bits $\rightarrow$ 8 bits

The textbook describes an overhead of:

25%

Trade-Off

Base64 makes binary data transportable through an ASCII-based mail system at the cost of additional data.

Quoted-Printable Encoding

Quoted-printable encoding is useful when:

- Most characters are ASCII.
- Only a small portion is non-ASCII.

Rules:

- ASCII character:



- Non-ASCII character:

= + Two Hexadecimal Digits

Quoted-Printable Example

Suppose a non-ASCII byte has hexadecimal value:

9D

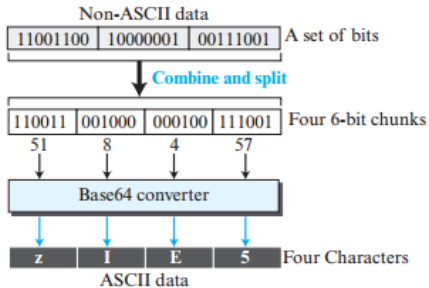
It is replaced by three ASCII characters:

=9D

Therefore:

Non-ASCII Byte $\rightarrow$ = + 9 + D

Figure 2.27 Base64 conversion



Content-ID and Content-Description

Content-ID

Uniquely identifies the complete message in a multiple-message environment.

Content-Description

Describes the nature of the message body.

Examples:

- Image
- Audio
- Video

Web-Based Mail

Some websites provide e-mail services through a web browser.
The textbook examples include:

- Hotmail
- Yahoo
- Google Mail

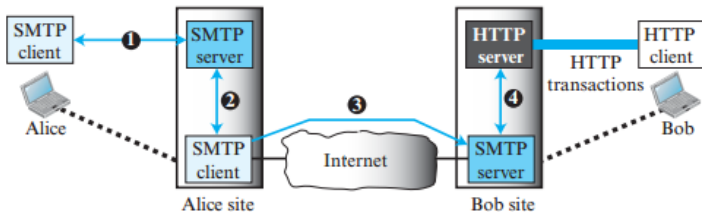


Web-based mail combines:

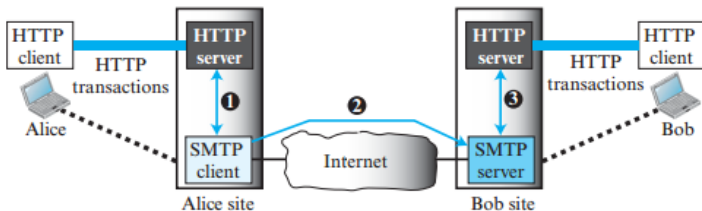
- HTTP
- SMTP

Two common cases are possible.

Figure 2.29 *Web-based e-mail, cases I and II*



Case 1: Only receiver uses HTTP



Case 2: Both sender and receiver use HTTP

Web-Based Mail: Case I

In Case I:

- Alice uses a traditional mail server.
- Bob uses a web-based mail server.

The communication path is:

Alice $\xrightarrow{\text{SMTP}}$ Alice's Mail Server

Alice's Server $\xrightarrow{\text{SMTP}}$ Bob's Web Mail Server

Bob's Server $\xrightarrow{\text{HTTP}}$ Bob's Browser

Case I: Bob Retrieves Mail

When Bob wants to retrieve his e-mail:

- ① Bob's browser sends an HTTP request.
- ② The website sends a login form.
- ③ Bob enters:
 - Login name
 - Password
- ④ The server verifies the credentials.
- ⑤ The e-mail list is sent in HTML format.
- ⑥ Bob uses further HTTP transactions to read messages.

Important

HTTP replaces POP3 or IMAP4 between the web server and Bob's browser.

Web-Based Mail: Case II

In Case II:

- Alice uses web-based mail.
- Bob also uses web-based mail.
- They do not need to use the same provider.

Alice sends the message to her web server using:



HTTP

The two mail servers exchange the message using:



SMTP

Bob retrieves the message using:



HTTP

Case II: Protocol Flow

The complete protocol flow is:

Alice's Browser $\xrightarrow{\text{HTTP}}$ Alice's Web Mail Server

Alice's Mail Server $\xrightarrow{\text{SMTP}}$ Bob's Mail Server

Bob's Mail Server $\xrightarrow{\text{HTTP}}$ Bob's Browser

Key Point

Even when both users use web-based mail, SMTP is still used between the mail servers.

The basic e-mail protocols discussed here do not provide security by themselves.

Application-layer security can be added using:

- 1 **Pretty Good Privacy (PGP)**
- 2 **Secure/Multipurpose Internet Mail Extensions (S/MIME)**

These mechanisms can provide protection for e-mail communication.

Electronic Mail: Key Takeaways

- E-mail uses intermediate mail servers.
- UA helps users compose, send, and read mail.
- MTA transfers messages using SMTP.
- SMTP is a push protocol.
- POP3 and IMAP4 are pull protocols.
- POP3 is simple but limited.
- IMAP4 provides advanced server-side features.
- MIME allows non-ASCII and multimedia content.
- Web-based mail combines HTTP and SMTP.
- PGP and S/MIME can provide e-mail security.